

The Icarus costume is not a hop-1 identity switch

A note on the fly_icarus chain. Not a result. Locks stay on the trees.

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Index of locks: <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

Locks stay on the trees. Index: gist 12835f74. Object of study for fly_icarus is constrained toy dynamics on named types. Hop-1 trees from fly_p1_sign onward are connectome measurements. Falsifier on the 11-cell slice: condition 3 not matching condition 1 when HD-on / cVA-off.

Abstract

Question: does a male P1 slice treat a frozen Icarus-from-male as a female? Icarus-from-male keeps male wiring. The object is fat, wingless, and grounded.

On an 11-cell published-sign row, condition 3 matches condition 1 when 7,11-HD is on and cVA is off. Schema w_crit = -1.5262. Template w[P1, DA1] = -1.8. @2cf5fd6.

Hop-1 signed counts onto 88 pC1-coexpress cells: LC10a +606, DA1 +62 ACh, HD -3, ppk23 0. hd_on_cva_off_ns is false. 3 tracks 1 on LC10a. @45aa064.

mAL GABA onto those 88 is -8810. Folded onto the cVA slot, that dump is past w_crit. DA1 hop-1 onto mAL_m* is 0. Numbers @e16856c. Sentence @41437dc.

DA1 reaches 90 of the 93 mAL GABA IDs at hop-2 through type-LH (8773 of 8810). Hop-1 DA1 onto those IDs is 3 of 8810. @de95257.

On the 161 LH cells in that hop, f-bar is 0.099 of all input and 0.703 of uniglomerular PN input. LHAV4c2 carries 1012 of 3301 LH-to-mAL synapses with 5 DA1 synapses. @01155c3.

--unfreeze, --female-brain-icarus, and --n 1000 stay stubbed.

Keywords

Drosophila melanogaster; P1; Icarus; cVA; 7,11-HD; LC10a; mAL; MaleCNS; hop-1; hop-2.

1. Question

Does a male P1 slice treat a frozen Icarus-from-male as a female?

Icarus-from-male keeps male wiring. The object is fat, wingless, grounded.

2. Locks

Schema. 11-cell published-sign row. 3 matches 1 when 7,11-HD is on and cVA is off. Fat, wingless, pinned contact, and female cuticle leave DA1 in place. Schema w_crit = -1.5262 on the 3d pin. Template w[P1, DA1] = -1.8. Song bouts = 1: latch on at step 36, dt = 0.05. @2cf5fd6, logs/p1_contact_s1.json, logs/p1_da1_dose_s1.json.

Hop-1 map. Same battery, MaleCNS hop-1 onto 88 pC1-coexpress cells. LC10a +606. DA1 +62 acetylcholine. HD -3. ppk23 0. ORN_DA1 to DA1 PN is +32,016. Direct ORN to P1 is 0. Every row sings. 3 tracks 1 (d31 = 0.0) because both objects drive LC10a. 3b P1 is 0.9421, above condition 1. hd_on_cva_off_ns is false. Seeds 2 and 3 match. @45aa064, logs/p1_sign_s1.json.

mAL dump. Hop-1 GABA from type-mAL onto those 88 is -8810. Folded into w[P1, DA1] on the parent LC10a scale, the cVA slot is -25.9842, past parent w_crit = -1.539. DA1 hop-1 onto mAL_m* is 0; onto two mALB1 cells it is 96 ACh. hd_on_cva_off_ns is true on that folded row because the slot was given -25.9842. Numbers @e16856c, logs/p1_mal_s1.json. Sentence @41437dc.

Drive. The 93 mAL GABA cells that make the -8810: DA1 hop-1 hits one mALB1 (45 ACh, 3 of 8810). Type-LH hop-1 hits 93 of 93. DA1 onto type-LH that hit those 93: 161 cells, 8425. Those 161 onto the 93: 3301, covering 90 cells and 8773 of 8810. Hop: DA1_PN -> type-LH -> mAL_GABA_pre. On this tree, DA1-driven means any DA1 synapse (f_i > 0). @de95257, logs/drive_s1.json.

Fraction. Frozen 161 IDs from @de95257. Count in: DA1 onto those 161 is 8425. Count out: those 161 onto the 93 is 3301. Incoming weight by pre tag: other 155,526, LH 72,266, other_uPN 33,072, multi_PN 19,140, DA1_PN 8,425, mAL 4,234. f-bar = 0.099172 (all incoming). f-bar^PN = 0.702674 (uniglomerular PNs). LHAV4c2: 2 cells, 1012 of 3301, 5 DA1 synapses, f-bar 0.002537, f-bar^PN 1.0 (5/5). LH007m: 814 of 3301, 1714 DA1, f-bar^PN 0.728904. LH008m: 360 of 3301, 2851 DA1, f-bar^PN 0.73711. @01155c3, logs/lh_frac_s1.json.

3. Stubbed flags

--unfreeze, --female-brain-icarus, and --n 1000 stay stubbed on every tree in this chain.

4. Replay

Each lock is one extract plus, where a battery exists, the frozen-object assay.

Tree	Extract	Lock
fly_icarus	data/templates/male_p1.json	logs/p1_contact_s1.json, logs/p1_da1_dose_s1.json
fly_p1_sign	scripts/extract_p1_weights.py	logs/p1_sign_s1.json
fly_p1_mal	scripts/extract_p1_weights.py	data/templates/extract.json; do not restamp logs/p1_mal_s1.json
fly_mal_drive	scripts/extract_drive.py	logs/drive_s1.json
fly_lh_frac	scripts/extract_frac.py	logs/lh_frac_s1.json; IDs in data/frozen/ids.json

How to run: in each tree, .venv/bin/python -m pytest. Rebuilds, where allowed, are the extract scripts above. Do not overwrite the listed lock files.

5. Repo table

Tree	SHA	Allowed sentence	Lock file
fly_icarus	2cf5fd6	In this 11-cell slice, HD-on / cVA-off is necessary and sufficient for 3 matching 1. Morphology, pin, and feminized cuticle are not. Schema W_crit=-1.5262.	logs/p1_contact_s1.json
fly_p1_sign	45aa064	Hop-1 DA1 is +62 ACh. LC10a +606. ppk23 hop-1 is 0. hd_on_cva_off_ns is false.	logs/p1_sign_s1.json
fly_p1_mal	e16856c	Hop-1 mAL GABA onto these 88 is large enough to pass W_crit if assigned to cVA. DA1 hop-1 onto mAL_m* is 0; onto two mALB1 cells it is 96 ACh.	logs/p1_mal_s1.json
fly_p1_mal sentence	41437dc	Same allowed sentence as the row above.	README of fly_p1_mal
fly_mal_drive	de95257	Yes at hop-2. DA1 hop-1 is 3 of 8810. DA1-driven type-LH accounts for 8773 of 8810. Hop: DA1_PN -> type-LH -> mAL_GABA_pre.	logs/drive_s1.json

Tree	SHA	Allowed sentence	Lock file
fly_lh_frac	01155c3	f-bar=0.099172. f-bar^PN=0.702674. LHAV4c2 carries 1012 of 3301 with 5 DA1 synapses.	logs/lh_frac_s1.json

Figure 1 (figures/stack.png): object tags to the 11-cell row to hop-1 counts to hop-2 LH to the f table.

Accountability (2026-09-20)

Results. Locked metrics are Python engine output at the tree SHAs named in this note. They were not generated by a language model.

Text. This note was drafted with Grok (xAI). The depositor reviewed the text against the locked logs and takes responsibility for it. No AI tool is an author.

Cite this note (2026-09-20)

This file is a note, not a finding and not a software deposit. Cite the frozen slice on the tree:

Martial Systems LLC. (2026). fly_icarus published-sign slice (science-2cf5fd6) [Computer software]. https://github.com/martialsystems/fly_icarus/tree/science-2cf5fd6

SWHID: swh:1:snp:f0619c4c7ef5bc90965f630063cd254ce48bba58

There is no DOI. The gist Artifact DOI cell stays empty until a 10. identifier exists.

Sources (2026-09-20)

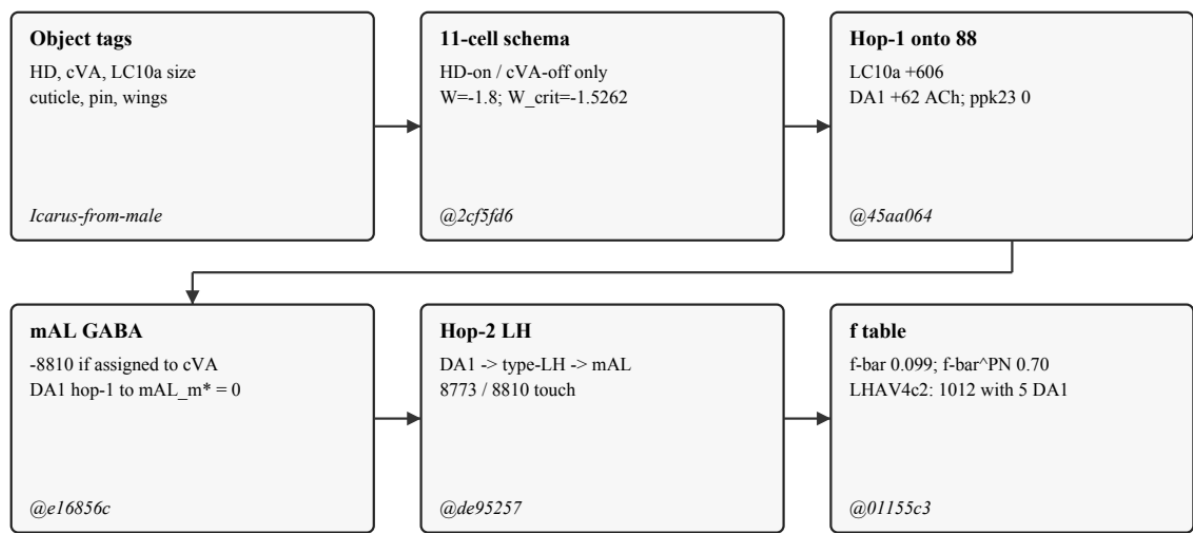
Records checked on Crossref, 2026-09-20. Female parent count is FlyWire. Male hop-1 counts are MaleCNS. The 11-cell row is a published-sign schema, not a hop-count extract.

Berg, S., Beckett, I. R., Costa, M., Schlegel, P., Januszewski, M., Marin, E. C., Nern, A., et al. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*, 189(18), 5504-5526.e15. <https://doi.org/10.1016/j.cell.2026.08.015>

Dorkenwald, S., Matsliah, A., Sterling, A. R., Schlegel, P., Yu, S. C., McKellar, C. E., Lin, A., et al. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634(8032), 124-138. <https://doi.org/10.1038/s41586-024-07558-y>

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Figure 1. Object tags to schema row to hop-1 counts to hop-2 LH to the f table (2026-09-16).



Locks stay on the trees. Pointer stack, not a new measurement.

Pointer stack across the locked trees. Not a new measurement.

Revisions (2026-09-16)

2026-09-16: first note from the locked trees. Same day: dropped the Not-stack; sentences taken from the lock files. Index gist 12835f74 left unedited.

2026-09-20: sources for MaleCNS (Berg et al., 2026) and FlyWire (Dorkenwald et al., 2024). Accountability sentence. Cite tag science-2cf5fd6. No DOI. This PDF is not a software deposit.